

Microeconomics Analysis: Search, Applications

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Motivation

We are going to look at three applications of search theory to consumer markets:

- Basic model of sequential search and oligopoly – Diamond (1970)
- Fixed sample search and price dispersion – Burdett and Judd (1983)
- Model of a search engine – Eliaz and Spiegler (2011)

Motivation

Search frictions play an important role in consumer markets.

There are two types of information consumers are trying to learn:

- information about products: availability, how much they like a particular product
- information about prices

Search goods are different from experience goods:

- for search goods consumers learn full information upon visiting the seller
- for experience goods consumers learn full information upon consuming the product

As we will see soon even tiny search friction can dramatically affect market outcomes

Diamond (1971): Setting

- n firms sell homogenous good and compete in prices
- normalize cost of production to zero
- unit mass of consumers, all value product at $v > 0$
- first search is free ("local store"), then $s > 0$ per firm searched
- search is sequential and random

If there were no search costs, we would have Bertrand competition with $p_i = 0$

This is not an equilibrium with $s > 0$: a firm can deviate by charging $p_i = s$, so consumers buy immediately (or $p_i = s - \varepsilon$)

Diamond (1971): Analysis

Is there a symmetric pure strategy equilibrium with $p^* < v$?

No, deviation to $p = \min\{p^* + s, v\}$ is profitable

Setting $p^* = v$ is an equilibrium:

- 1 as all firms charge v , consumers have no incentives to search
- 2 as consumers do not search, the optimal price is $p^* = v$

What about mixed strategies $F(p)$? Profitable deviation: $E(p) + s$.

Diamond (1971): Key Points

- 1 Even a tiny search friction flips outcome from competitive to monopoly pricing
- 2 If the first visit was costly things would be even worse
- 3 In this case market would collapse, due to hold-up problem
- 4 Downward sloping demand leaves consumers with some surplus at monopoly price, but the market does not need to collapse
- 5 Probably predictions are too stark to be realistic...

Burdett and Judd (1983): Setting

We start with a **fixed sample** search model by Burdett and Judd (1983)

- n firms sell homogenous good and compete in prices
- normalize cost of production to zero
- unit mass of consumers, all value product at $v > 0$
- consumers decide how many firms they want to sample
- cost is s per firm
- consumers buy at the lowest price, provided that is less than v

Burdett and Judd (1983): Preliminaries

Clearly there is a Diamond-style equilibrium with $p^* = v$ and no search

Pure strategy symmetric equilibria with $p^* < v$ do not exist:

- if all consumers sample not more than one price, then firms are better off by deviating to $p = v$
- if some consumers sample two or more firms with positive probability, then a firm can do better by charging $p^* - \varepsilon$ and capturing all these consumers

Thus, we are left with mixed strategies. Denote the equilibrium strategy as $F(p)$

Burdett and Judd (1983): Search

Denote probability that a consumer samples k firms as q_k

Claim 1: $q_1 > 0$. If $q_1 = 0$, then all consumers participating in the market compare at least two firms, which leads to $p^* = 0$. However, if $p^* = 0$ then sampling one firm is optimal.

Claim 2: $q_1 < 1$. If $q_1 = 1$, then all consumers participating in the market draw just one quote, thus firms should set $p^* = v$ so either (i) market collapses or (ii) assuming first search is free – monopoly pricing equilibrium.

Total cost from searching k products and buying at the lowest price:

$$sk + \int_{\underline{p}}^{\bar{p}} kp[1 - F(p)]^{k-1} dF(p)$$

is convex in k , this at most two adjacent q 's can be positive $\Rightarrow q_2 > 0, q_1 + q_2 = 1$

Burdett and Judd (1983): Profit and Pricing

Standard arguments: $F(p)$ is continuous with convex support

Denote $q_1 = q$, $q_2 = 1 - q$. Profit function:

$$\pi = \frac{1}{n} p [q + 2(1 - q)(1 - F(p))]$$

Boundary condition: $\bar{p} = v$

This gives

$$F(p) = 1 - \frac{q}{2(1 - q)} \frac{v - p}{p}, \quad p \in \left[\frac{qv}{2 - q}, v \right]$$

$$\pi = \frac{qv}{n}$$

Note, that total market profit is qv , which is independent of n . Thus, having more firms does not reduce the average price paid.

Lower search activity, i.e. higher q increase price paid due to: shift of price distribution and smaller number of price comparisons.

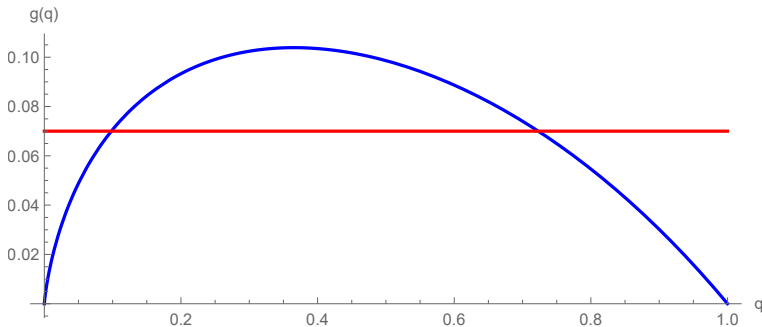
Burdett and Judd (1983): Search in Eqm

Indifference condition:

$$s + \int_{qv/(2-q)}^v p dF(p) = 2s + \int_{qv/(2-q)}^v p d[1 - (1 - F(p))^2]$$

which gives

$$g(q) \equiv \frac{q(\ln(\frac{2-q}{q}) - 2(1-q))}{2(1-q)^2} = \frac{s}{v}$$



Burdett and Judd (1983): Takeways

Equilibrium with search exists if search frictions are sufficiently low

There is price dispersion in equilibrium

Consumers randomize between sampling one and two firms

Search activity benefits consumers in two ways: they more often draw the best of two offers and the price distribution is improved

There is a Diamond collapsed market equilibrium

Eliaz and Spiegel: Firms

- The model is based on Eliaz and Spiegel (EJ, 2011)
- A large (infinite) number of firms producing differentiated products
- Firms reach their customers via a monopolistic search engine
- Firms are vertically differentiated w.r.t. the quality (relevance): $\theta \in [0, 1]$ which is distributed according to $G(\theta)$
- If firm i 's relevance is θ_i , a consumer obtains match value 0 with probability $1 - \theta_i$, and match value v with probability θ_i , where v is randomly drawn from a common distribution F on $[0, 1]$ – a horizontal part of product differentiation

Eliaz and Spiegler: Illustration

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Eliaz and Spiegel: Search engine

- A monopoly search engines controls the quality of the search pool
 - It chooses quality level θ^* so only firms with $\theta_i \geq \theta^*$ appear in the search pool
- The search engine does not need to directly control θ^* , but can implement any θ^* by setting per-click fee r
 - this is a payment from firms to the search engine each time a consumer visits the firm, which is independent of whether the transaction is successful or not
 - only firms that accept the per-click fee are allowed to enter the search pool
- Only firms with high enough quality will enter the pool since their expected profits are high enough to compensate the payment to the search engine

Eliaz and Spiegel: Timing

- Timing
 - the search engine sets r
 - firms decide whether to accept r and enter the search pool
 - consumers search sequentially at cost s using the search engine and make transactions
 - the search engine collects payments from firms
- We solve the model by backward induction
 - first, take the set of firms that enter the search pool as given, solving for the equilibrium price p^*
 - incorporate this characterization into the search engine's problem, and determine the optimal per-click fee r^*

Eliaz and Spiegler: Consumer search

- Only firms with the relatively high relevance will enter the search pool. Define the cut-off type as θ^* and only firms with $\theta \geq \theta^*$ enter the search pool; denote

$$\theta^e = \mathbb{E}(\theta \mid \theta \geq \theta^*)$$

- Consumers' optimal stopping rule is determined by the reservation price v^*

$$\theta^e \int_{v^*}^1 (z - v^*) dF(z) = s$$

Derivation

A consumer, who expects equilibrium price p^* , stops and buys if $v - p \geq v^* - p^*$ and continues on searching if otherwise

- The term in the front is the conditional expectation of relevance. Consumers take into account that a match is of positive value only with average probability θ^e

Eliaz and Spiegel: Firm pricing

- In firms' pricing problem, first thing to notice is that the per-click fee does not affect the pricing decision directly
- A firm has to pay r once a consumer clicks in, no matter whether the transaction is successful or not, and, thus, r is taken as sunk
- A firm with quality θ solves the following problem

$$\max_p p\theta(1 - F(v^* - p^* + p)),$$

which yields

$$p^* = \frac{1 - F(v^*)}{f(v^*)}$$

Eliaz and Spiegler: profit and search duration

- A type- θ firm's em profit is

$$\pi(\theta) = p^* \theta (1 - F(v^*)) = \frac{\theta (1 - F(v^*))^2}{f(v^*)}$$

- Equilibrium expected conversion rate is $\theta^e (1 - F(v^*))$
- Equilibrium expected search duration is the inverse of the expected equilibrium conversion rate, $1/[\theta^e (1 - F(v^*))]$

Eliaz and Spiegler: Search Engine Fee

- Turn to the search engine's optimization problem
- A firm will enter the search pool when its expected profit per-click exceeds the per-click fee

$$\frac{\theta(1 - F(v^*))^2}{f(v^*)} \geq r$$

- The cut-off θ^* is determined by an indifferent firm

$$\frac{\theta^*(1 - F(v^*))^2}{f(v^*)} = r,$$

and all firms with $\theta \geq \theta^*$ will enter

Eliaz and Spiegler: Search engine Profit Maximization

- The search engine maximizes its expected profit, which equals the per-click price times the expected number of clicks (expected search duration)

$$\max_r \frac{r}{\mathbb{E}[\theta^e(1 - F(v^*))]}$$

- Plug in the expression of r from previous slide, the maximization problem becomes

$$\max_{\theta^*} \frac{\theta^*}{\theta^e} \frac{1 - F(v^*)}{f(v^*)}$$

Eliaz and Spiegler: Firm quality and per-click fee

- θ^e increases in θ^*
- First, consider the equilibrium per-click fee

$$r^* = \text{conv. rate} \times p^* = \theta^*(1 - F(v^*)) \times \frac{(1 - F(v^*))}{f(v^*)}$$

- Increasing θ^* has an ambiguous effect on equilibrium per-click fee
 - θ^* increases $\Rightarrow v^*$ increases $\Rightarrow p^*$ decreases
 - θ^* increases $\Rightarrow$ conversion rate can go either way

- Next consider search duration

$$\frac{1}{\theta^e[1 - F(v^*)]}$$

- Increasing θ^* has an ambiguous effect on search duration
 - θ^* increases $\Rightarrow \theta^e$ increases
 - θ^* increases $\Rightarrow 1 - F(v^*)$ decreases

Eliaz and Spiegel: Search distortion

Explanation:

- On search duration:
 - when the marginal quality increases, consumers are expected to be matched more quickly, which reduces the expected search duration
 - when the marginal quality increase, conducting additional search becomes more tempting. Consumers become more “picky” and increase their reservation price v^* . This effect extends the expected search duration
- On per-click price:
 - when the marginal quality increases, the competition between firms becomes fiercer as consumers are more picky now, thus lowers product price
 - however, the conversion rate can go either way due to the same reason we stated for the search duration

Eliaz and Spiegler: Search distortion

- We can also rearrange the search engine's equilibrium profit function π^* in the following way

$$\frac{\pi^*}{p^*} = \frac{\theta^*}{\theta^e}$$

- The L.H.S. can be understood as unit-free measure of the intermediation fee that the search engine extracts
- The R.H.S. is the ratio between the relevance of the marginal and the average firms in the search pool. When there is no distortion, this ratio equals 1
- Since the effect of increasing θ^* is ambiguous, it is not a priori clear whether the search engine wants to distort the relevance
- However, for our purpose, it suffices to show that the distortion can happen for some distribution functions $F(v)$ and $G(\theta)$

Stopping Rule

Take a look at a consumer who is indifferent between buying and searching. At what v^* does this happen?

$$\begin{aligned} v^* = & -s + \Pr(v = 0) \cdot v^* + \Pr(v \neq 0) [\Pr(v < v^*)v^* + \\ & \Pr(v \geq v^*)\mathbb{E}(v|v \geq v^*)] = \\ & -s + [1 - \mathbb{E}(\theta | \theta \geq \theta^*)]v^* + \mathbb{E}(\theta | \theta \geq \theta^*) \left[F(v^*)v^* + \int_{v^*}^1 z dF(z) \right] \end{aligned}$$

Rearranging gives:

$$\begin{aligned} 0 = & -s + \mathbb{E}(\theta | \theta \geq \theta^*) \left[-(1 - F(v^*))v^* + \int_{v^*}^1 z dF(z) \right] = \\ & -s + \mathbb{E}(\theta | \theta \geq \theta^*) \int_{v^*}^1 (z - v^*) dF(z) \end{aligned}$$